Transformations of Functions

Notes with Similar Practice Problems

Transformations

A **transformation** changes the position, shape, or orientation of a graph.
There are several major types of transformations:

- Horizontal Shifts
- Vertical Shifts
- Horizontal Stretches and Shrinks
- Vertical Stretches and Shrinks
- Reflections

1. Horizontal Shifts

$$y = f(x - h)$$

moves the graph:

right h units

$$y = f(x + h)$$

moves the graph:

left h units

2. Vertical Shifts

$$y = f(x) + k$$

moves the graph upward by:

k units

$$y = f(x) - k$$

moves the graph downward by:

k units

3. Horizontal Stretch and Shrink

$$y = f(ax)$$

causes a horizontal shrink by a factor of:

$$\frac{1}{a}$$

The x-coordinates divide by a.
If:

$$(x, y)$$

is on the graph of:

$$y = f(x)$$

then:

$$\left(\frac{x}{a}, y\right)$$

is on the graph of:

$$y = f(ax)$$

Similarly,

$$y = f\left(\frac{x}{a}\right)$$

causes a horizontal stretch by a factor of:

$$a$$

4. Vertical Stretch and Shrink

$$y = af(x)$$

causes a vertical stretch by a factor of:

$$a$$

The y-coordinates multiply by a.

$$y = \frac{1}{a}f(x)$$

causes a vertical shrink by a factor of:

$$\frac{1}{a}$$

5. Reflections

$$y = -f(x)$$

reflects the graph across the x-axis.

$$y = f(-x)$$

reflects the graph across the y-axis.

Summary

Most Important Ideas

Inside changes affect x-values inversely

Outside changes affect y-values directly

Horizontal Transformations

$$f(ax)$$

means:

$$x \to \frac{x}{a}$$

- $f(2x)$: horizontal shrink by $\frac{1}{2}$
- $f(3x)$: horizontal shrink by $\frac{1}{3}$
- $f\left(\frac{x}{2}\right)$: horizontal stretch by 2
- $f\left(\frac{x}{5}\right)$: horizontal stretch by 5

Vertical Transformations

$$af(x)$$

means:

$$y \to ay$$

- $2f(x)$: vertical stretch by 2
- $5f(x)$: vertical stretch by 5
- $\frac{1}{2}f(x)$: vertical shrink by $\frac{1}{2}$

Coordinate Mapping Rules

$$y = f(ax)$$

maps:

$$(x, y) \to \left(\frac{x}{a}, y\right)$$

$$y = af(x)$$

maps:

$$(x, y) \to (x, ay)$$

Key Principle

Any point satisfying the equation is on the graph, and every point on the graph satisfies the equation.

Practice Problems

1. Describe the transformation:

$$y = f(x - 4)$$

2. Describe the transformation:

$$y = f(x + 7)$$

3. Describe the transformation:

$$y = f(x) + 5$$

4. Describe the transformation:

$$y = f(x) - 3$$

5. Describe the transformation:

$$y = f(2x)$$

6. Describe the transformation:

$$y = f(5x)$$

7. Describe the transformation:

$$y = f\left(\frac{x}{2}\right)$$

8. Describe the transformation:

$$y = f\left(\frac{x}{6}\right)$$

9. Describe the transformation:

$$y = 3f(x)$$

10. Describe the transformation:

$$y = \frac{1}{2}f(x)$$

11. Describe the transformation:

$$y = -f(x)$$

12. Describe the transformation:

$$y = f(-x)$$

13. If $(8, 2)$ is on the graph of $y = f(x)$, find the corresponding point on:

$$y = f(4x)$$

14. If $(3, 9)$ is on the graph of $y = f(x)$, find the corresponding point on:

$$y = f\left(\frac{x}{3}\right)$$

15. If $(6, -1)$ is on the graph of $y = f(x)$, find the corresponding point on:

$$y = 2f(x)$$

16. If $(10, 5)$ is on the graph of $y = f(x)$, find the corresponding point on:

$$y = \frac{1}{2}f(x)$$

17. If $(4, 7)$ is on the graph of $y = f(x)$, find the corresponding point on:

$$y = -f(x)$$

18. If $(-3, 2)$ is on the graph of $y = f(x)$, find the corresponding point on:

$$y = f(-x)$$

19. Given the parent function:

$$y = x^2$$

graph:

$$y = f(2x)$$

and identify whether the graph is stretched or shrunk.

20. Explain why:

$$f(3x)$$

causes a horizontal shrink by a factor of:

$$\frac{1}{3}$$

using coordinate mapping.